Street Smart for ArcGIS Pro

User Manual · Version 3.0.26.4

PRODUCT

Street Smart for ArcGIS Pro

Add-in version 3.0.26.4

PREPARED BY

Cyclomedia Technology B.V.

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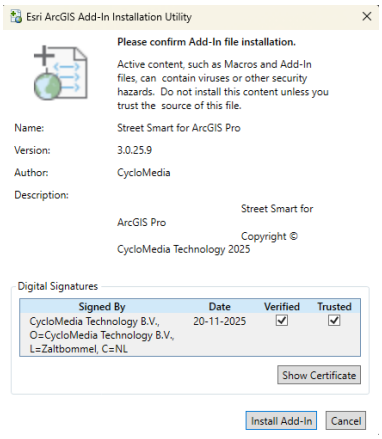


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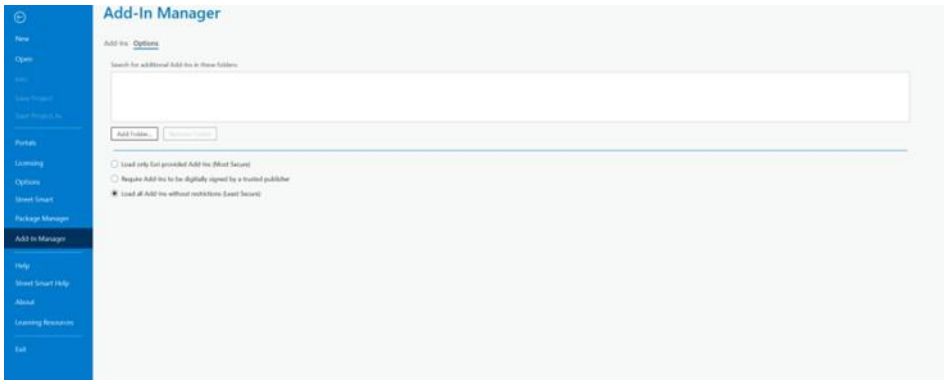
Installation

To install Street Smart for ArcGIS Pro, please perform the following steps:

1. Street Smart for ArcGIS Pro can be installed directly with the add-in file.
   - Double click the downloaded Esri add-in file.
   - In the opened Instalation Utility click on: Install Add-In.
   - After installation restart ArcGIS Pro.



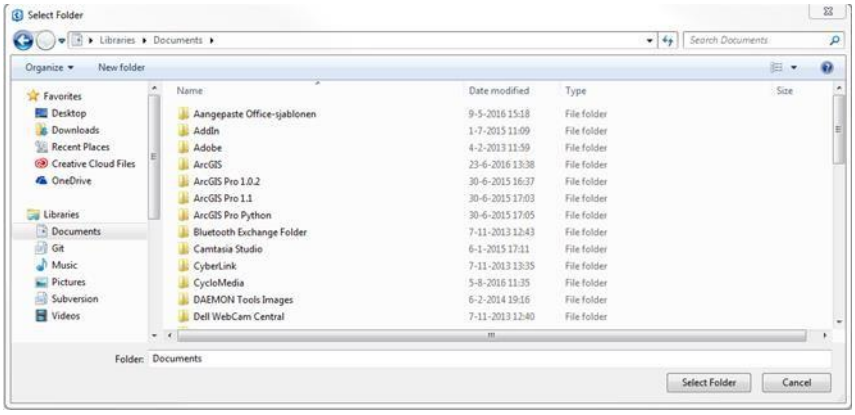
2. Alternatively, Street Smart for ArcGIS Pro can be installed from a disk or network location:
   - Copy the Add-In file to a (local) disk or network location.
   - Start ArcGIS Pro.
   - Select menu: *Project* ► *Add-In Manager* and select the *Options* tab.



Add-In Manager, Options tab.

- Click <*Add Folder*...>. Browse to the folder where the Add-In file is located and click

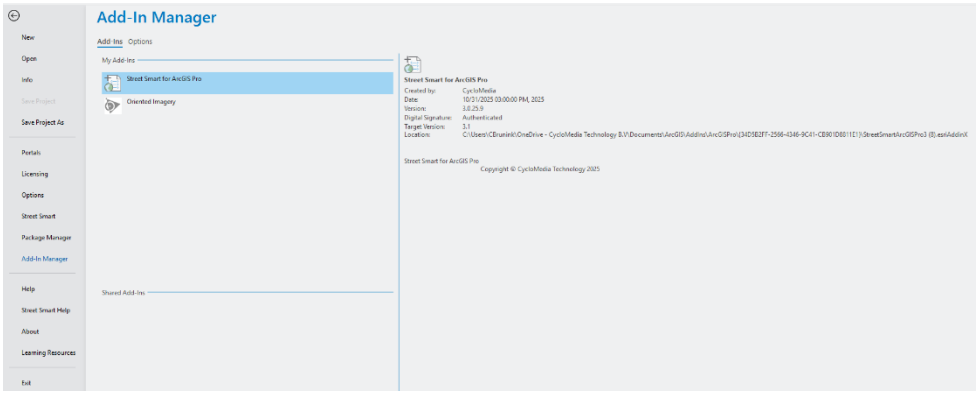
<*OK*>.



Browse to the folder where the Add-In file is located.

- Close the Add-In Manager and restart ArcGIS Pro.
- Select menu: *Project* ▶ *Add-In Manager*.
- Check whether Street Smart for ArcGIS Pro is '*Authenticated*'. If the Add-In is '*Untrusted*', close the Add-In Manager, close ArcGIS Pro and install the certificate

'thawte-intermediate-ca.cer' first (which is made available together with this user manual).



Add-In Manager, Add-Ins tab, check if Street Smart for ArcGIS Pro is 'Authenticated' or 'Untrusted'.

- To install the certificate, please perform the following steps:
  - Double-click on 'thawte-intermediate-ca.cer' to install the certificate.



Install the certificate.

- Click on <*Install Certificate...*> to start the installation wizard for the certificate.



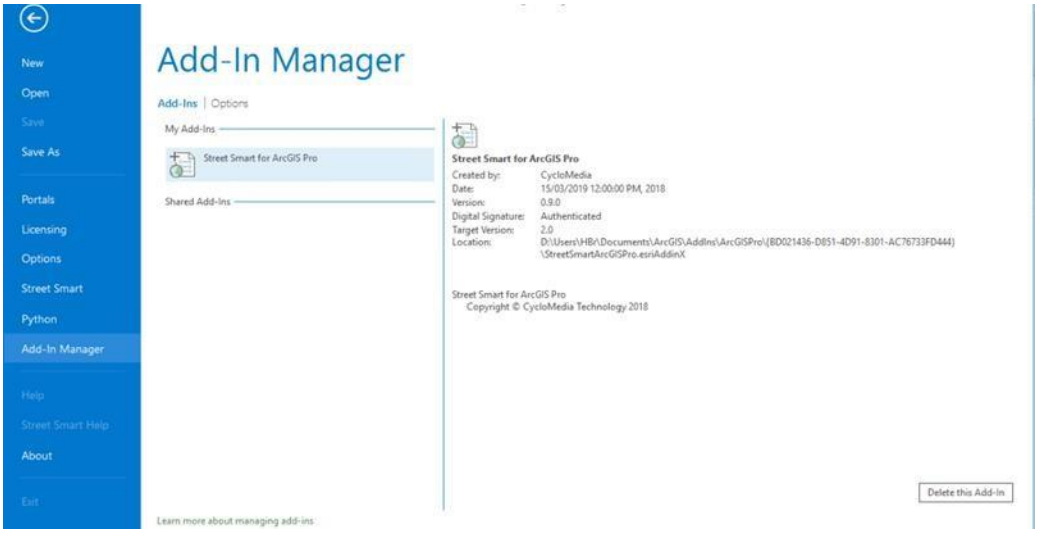
Certificate Import Wizard.

- Click: <*Next*>, <*Next*>, <*Finished*> to complete the installation of the certificate.



The certificate is successfully imported.

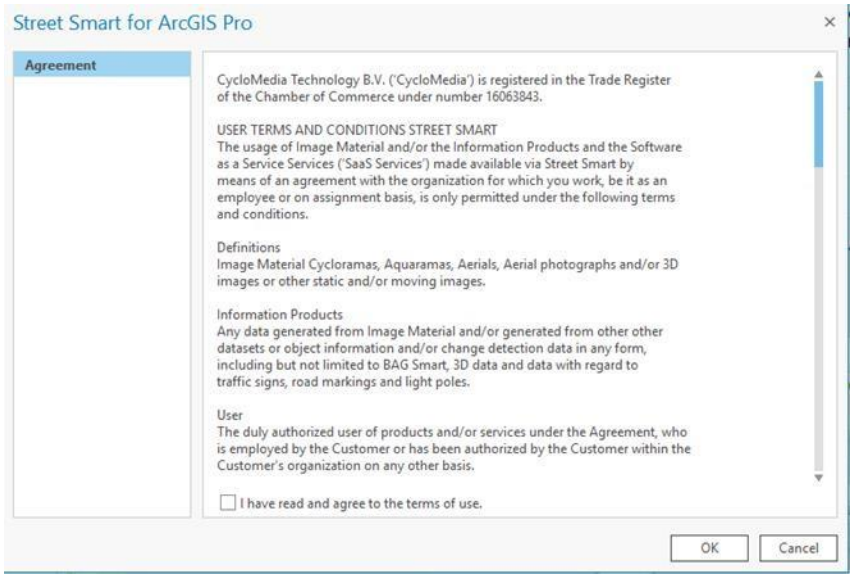
- Start ArcGIS Pro.
- Select menu: *Project* ► *Add-In Manager*.
- Check if Street Smart for ArcGIS Pro is now 'Authenticated'. You can then proceed to paragraph 2).



Add-In Manager, Add-Ins tab, check that Street Smart for ArcGIS Pro is now 'Authenticated'.

1. If you start ArcGIS Pro for the first time after installing the Add-In, you will be shown the agreement form. Mark '*I have read and agree to the terms of use*' as checked and click

<OK>.



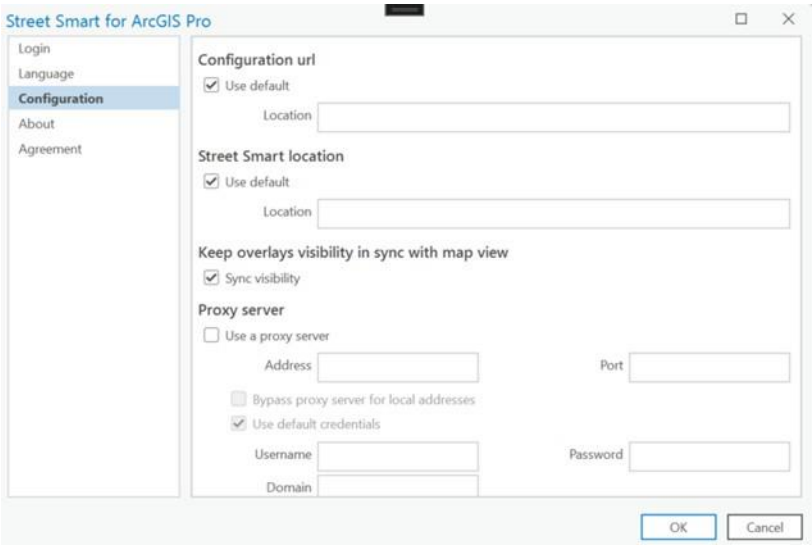
Agreement form.

2. Select the *Project* ▶ *Street Smart* menu. You will now open the window *Street Smart for ArcGIS Pro*.



Street Smart menu

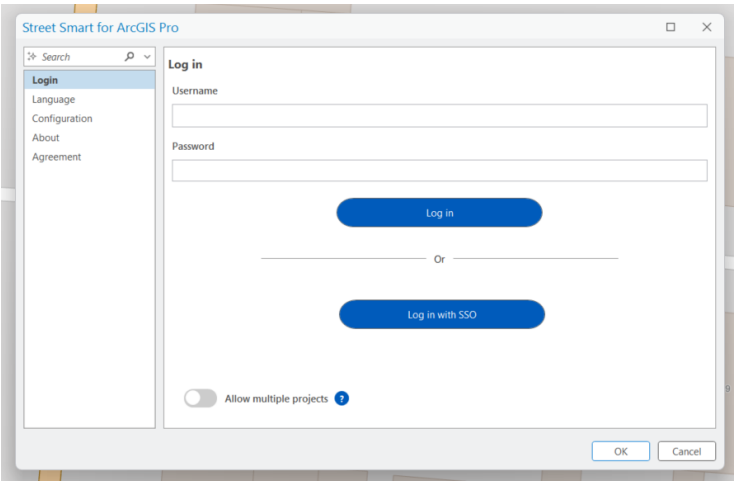
3. Select the tab *Configuration*.
4. If you use a proxy server, you can enable *Use a proxy server* and enter the data about your proxy server.



Street Smart for ArcGIS Pro Add-In window, Configuration tab.

5. If you are using a local installation, uncheck the base URL and the swf URL *Use default* option and enter the correct URLs at *Location* instead.

6. Select the *Login* tab.

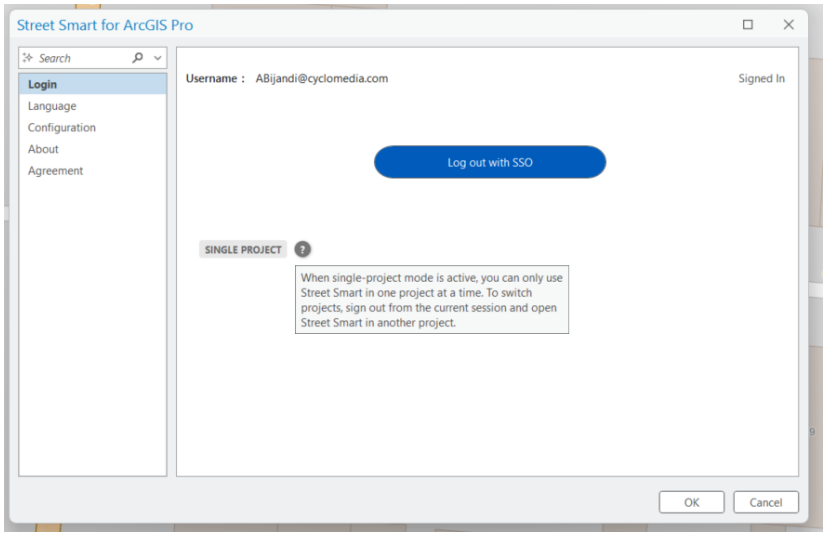


7. Now you can enter a Username and Password and click on button Log in to log in using Basic Authentication, or you can click "Log in with SSO" to log in using Single Sign On.

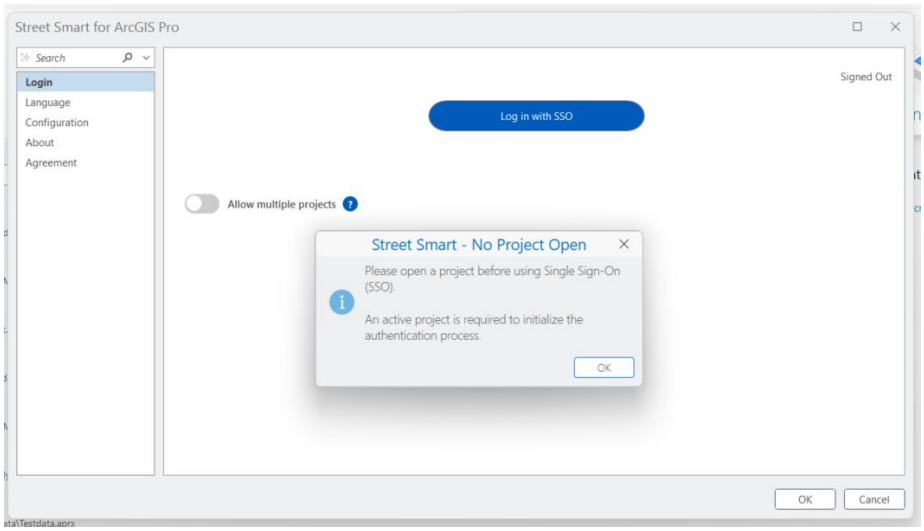
   – If the user chooses Basic Authentication and tries to log in using a username and password, they will receive a 'Login failed' notification if the username and/or password is incorrect. However, if everything is entered correctly, a new window will open, and they will receive a 'Login successful' notification

   – The Login tab also contains an **Allow multiple projects** toggle. When disabled (default), Street Smart operates in **Single Project** mode — only one ArcGIS Pro project can use the

add-in at a time. To switch projects, you must first sign out and then open Street Smart in another project. When enabled, Street Smart operates in **Multiple Projects** mode, allowing the add-in to run across multiple ArcGIS Pro projects simultaneously. Note that SSO with two-factor authentication is required for each individual project session when this mode is active.



Note: When using SSO, an active ArcGIS Pro project must be open before clicking *Log in with SSO*. If no project is open, a warning dialog will appear prompting you to open a project first.

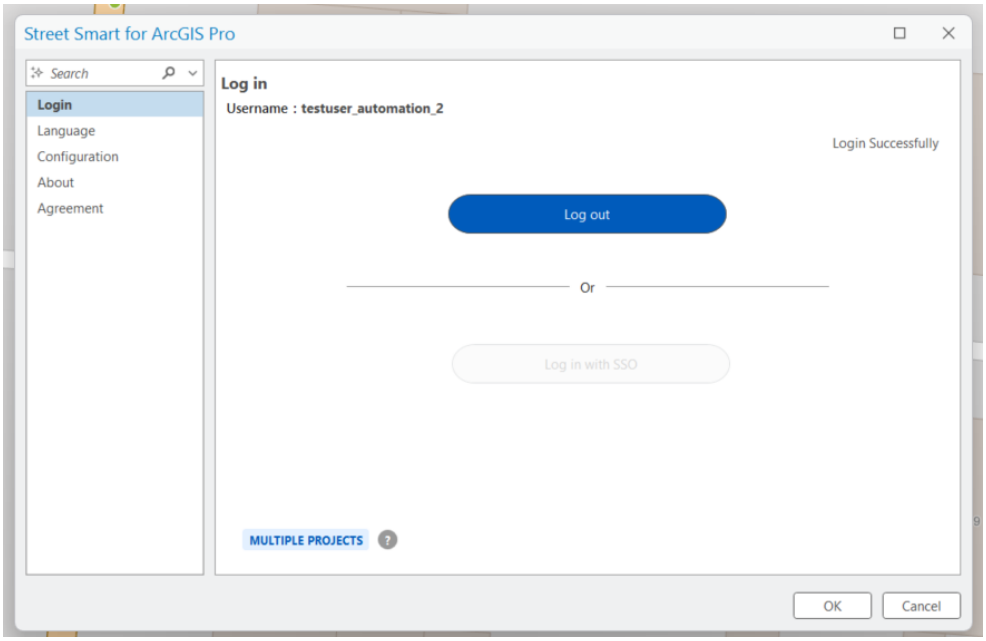


- If the user chooses Single Sign On (SSO) and tries to log in using the 'Log in with SSO' button, the Add-In will redirect them to the configured OAuth provider.
- And then the user will be logged in and granted permissions to the Add-In.



Login Form Behavior – Basic Authentication vs. Single Sign On & Multiple Instances

- When Single Sign On is selected as the authentication method, the login form only shows the relevant fields (e.g., Single Sign On button, status message).
- If Single Sign On is enabled, the fields for Basic Authentication (e.g., username, password) are hidden, providing a clean and intuitive user experience.
- The UI dynamically adjusts based on the user's login method selection.
- The Allow multiple projects toggle controls how many ArcGIS Pro projects can run Street Smart simultaneously. By default it is off (Single Project mode), meaning only one project can use the add-in at a time sign out first before switching to another project. When on (Multiple Projects mode), the add-in runs across several projects at once, with each session requiring its own SSO authentication.
- The active mode (SINGLE PROJECT or MULTIPLE PROJECTS) is shown on the Login tab after signing in. To switch modes, sign out, adjust the toggle, and sign back in.



- When logged in using Basic Authentication, the Single Sign On with SSO button will be visible but disabled, ensuring clarity without allowing redundant authentication actions.

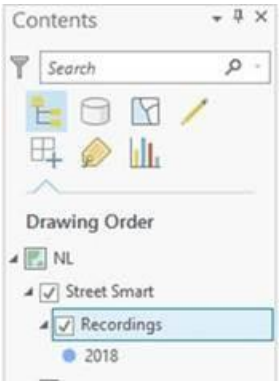
How to use Street Smart for ArcGIS Pro

– 2.1 Exploring and viewing Cycloramas in ArcGIS Pro

1. *How to show recording locations of Cycloramas*
    1. Select *Recent Recording Layer* in the Street Smart tab.

The selected Recent Recording Layer icon.

2. In the *Table of Contents* of ArcGIS Pro a new layer will appear as shown in the figure below. Zoom in on the map to a zoom level of at least 1:2000 to see the recording locations on the map (provided there are recordings in the area).

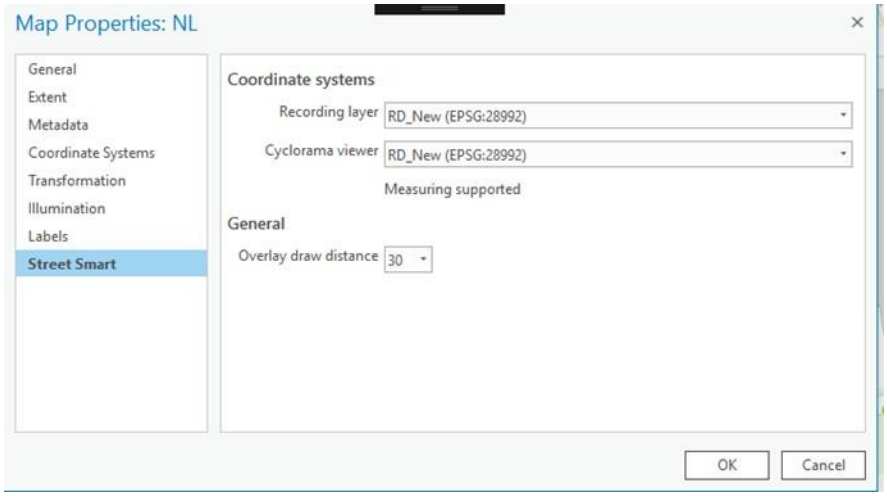


The layer with recordings in the Table Of Contents.

2. *How to configure the Cyclorama viewers*
    1. Select in the *Table of Contents* the map, right click on the mouse button ► *Properties*

menu. You will now open the window *Map Properties*.

1. Select the *Street Smart* tab.



Cyclorama viewing settings in the Street Smart tab.

2. Select the coordinate system which the viewer(s) should use.
3. Select the coordinate system which the Recording layer should use.
4. Select overlay draw distance of overlays (in meters)
5. Press <*OK*> for the changes to take effect.

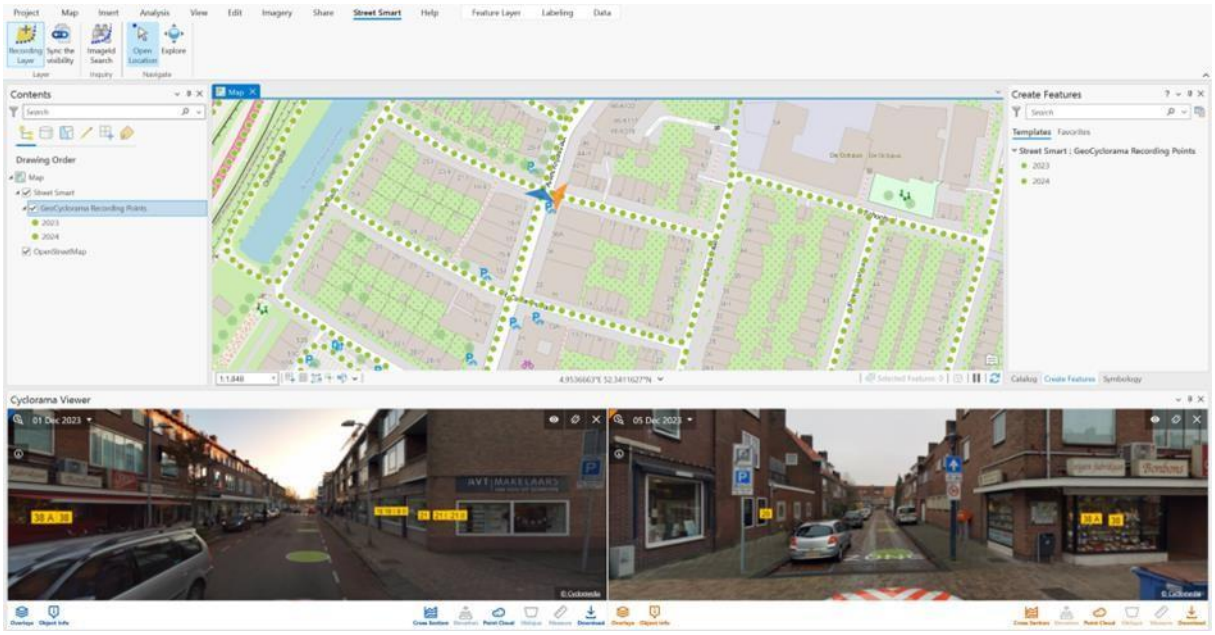
1. *How to open Cycloramas*

    1. Select the *Open Location Tool* in the Street Smart toolbar.



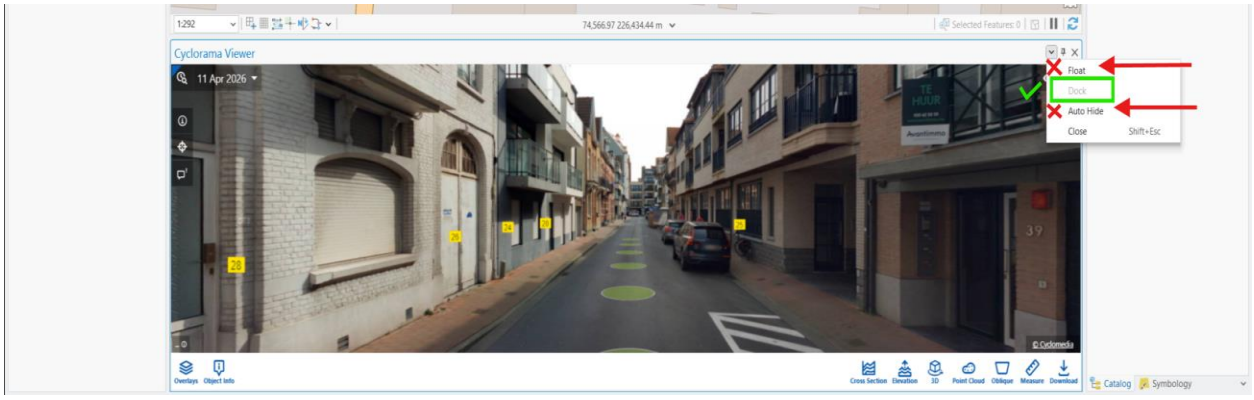
The selected Open Location Tool icon.

2. Click on a dot for the corresponding Cyclorama to open in a Cyclorama viewer.
3. Click on another dot with shift-pressed and a second Cyclorama will open.
4. If the shift key isn't pressed, the first Cyclorama will be replaced by the second Cyclorama.



Street Smart for ArcGIS Pro with two Cyclorama viewers opened.(shift-click)

5. ⚠ **Important: Cyclorama Viewer Window Settings**

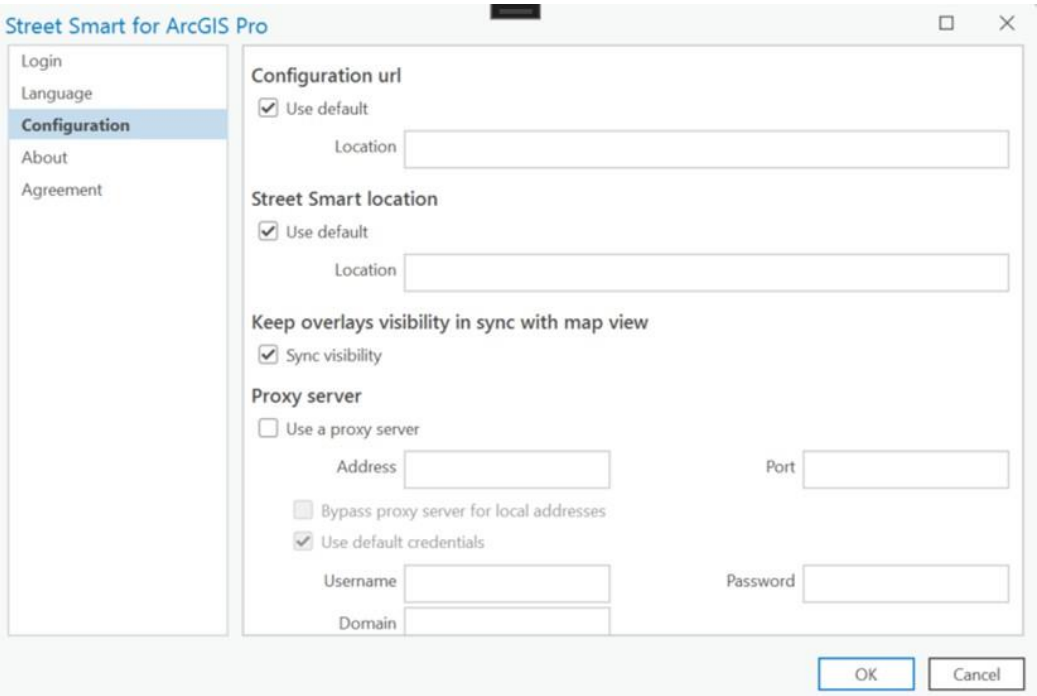


To ensure application stability, please keep the Cyclorama Viewer pane in the **Dock** position.

- Changing the pane setting to **Float** or **Auto Hide** may cause ArcGIS Pro to close unexpectedly or behave unreliably when logging in or managing the viewer.
- Please continue to use the **Dock** setting exclusively until a permanent resolution is provided in a future update.

– 2.2 Cyclorama Overlays Visibility Synchronization

New configuration option (Keep overlays visibility in sync with map view) has been added to allow users to control whether Cyclorama overlays stay in sync with the map view or not.



How it works

1. **By Default (Sync Enabled)**:
   - When a new project is opened, the overlays visible in Cyclorama will be the same as those visible in the map.
   - Users can change the visibility of overlays in Cyclorama, but if they close and reopen the project, the visibility will resync with the map unless desynced.

When Sync is Disabled (Checkbox Unchecked):

- Overlays will not automatically follow the map layer visibility.
- Once a user modifies the visibility of overlays, those preferences will be saved and remembered across project sessions.
- This desync may improve performance, particularly in projects with many layers, as only relevant overlays will be displayed.

Street Smart Toolbar Button (Sync the visibility):

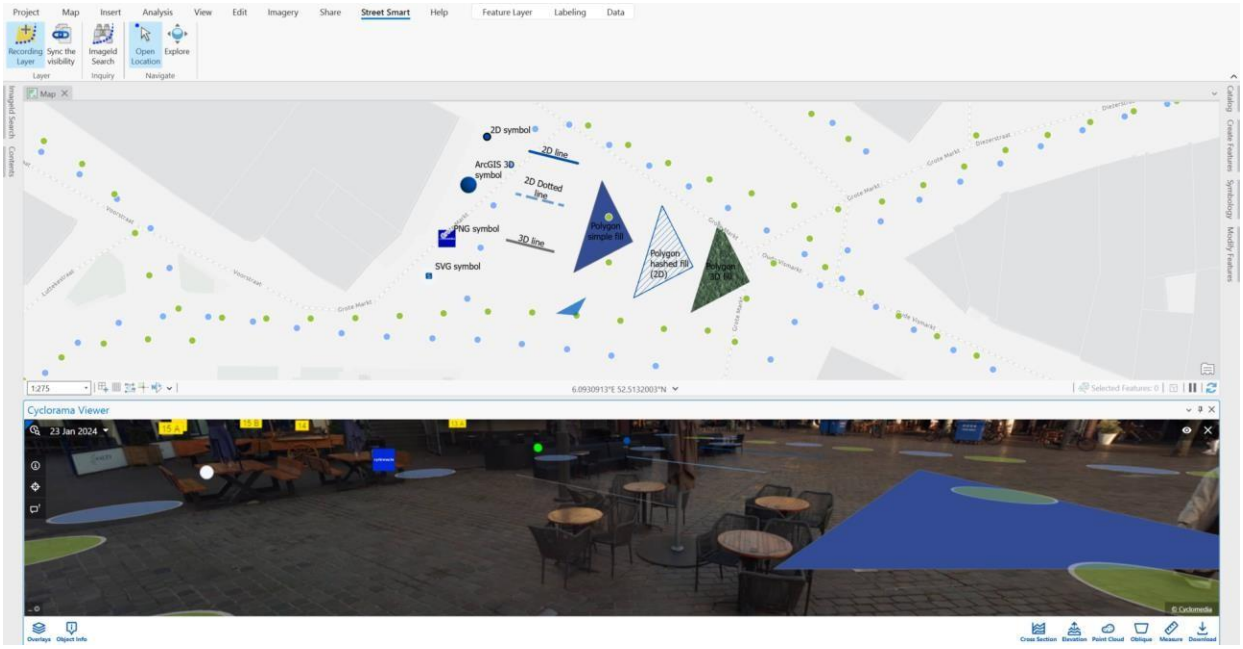
- An override button will be added to the **Street-Smart** toolbar, next to the **Recording layer**

button.

- This button allows users to override the synchronization setting at the project level.
- The override button will function as a shortcut, enabling or disabling the sync for that specific project, allowing for greater flexibility.

2.3 How to show vector data in Cycloramas

1. Add one of the various types of vector data to the Table of Contents, such as:
   - Shapefiles
   - ArcSDE
   - Personal MDB
   - File GDB
2. Open a Cyclorama. Note that the vector data will now be shown in the Cycloramas.



Showing vector data as overlay in the Cyclorama viewers.

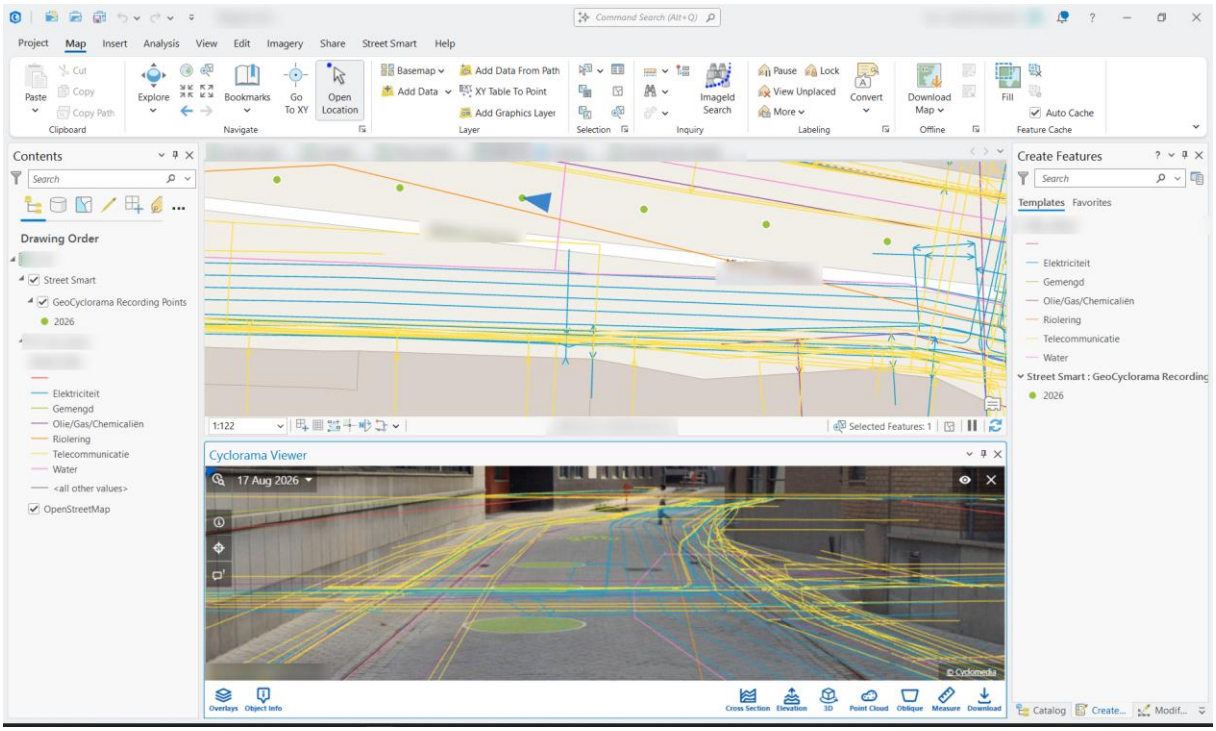
3. Optionally, set the distance at which vector layer data is displayed in the *Map Properties* ► *Street Smart* menu. The default distance is 30 meters, which means only vector data located within 30 meters of the recording location will be displayed.

⚠ **Note: Performance with large amounts of vector data**

The performance of the add-in has been improved in this release. Overlays are loaded and drawn in the Cyclorama viewer noticeably faster than in previous versions.

Keep in mind, however, that every vector layer that is switched on is projected onto each recording. If a single map contains a large amount of vector data for example a complete utility network with electricity, gas, sewerage, telecommunication and water lines expect some delay and slower response when a Cyclorama is opened, and while moving between recordings. The amount of data being drawn is the limiting factor here, not the add-in itself.

To keep the viewer responsive, switch off the layers you do not need before opening a Cyclorama. Only the layers that are visible in the map are drawn in the Cyclorama, so unchecking unneeded layers in the *Contents* pane directly reduces the work done at load time. The overlay draw distance described above has the same effect: a shorter distance means less data to draw. See also *2.2 Cyclorama Overlays Visibility Synchronization*.



An example of a map carrying a large amount of vector data. Every visible layer is projected onto each recording — which is what makes this view so valuable, and also what makes it heavier to draw.

2.4 Performing measurements

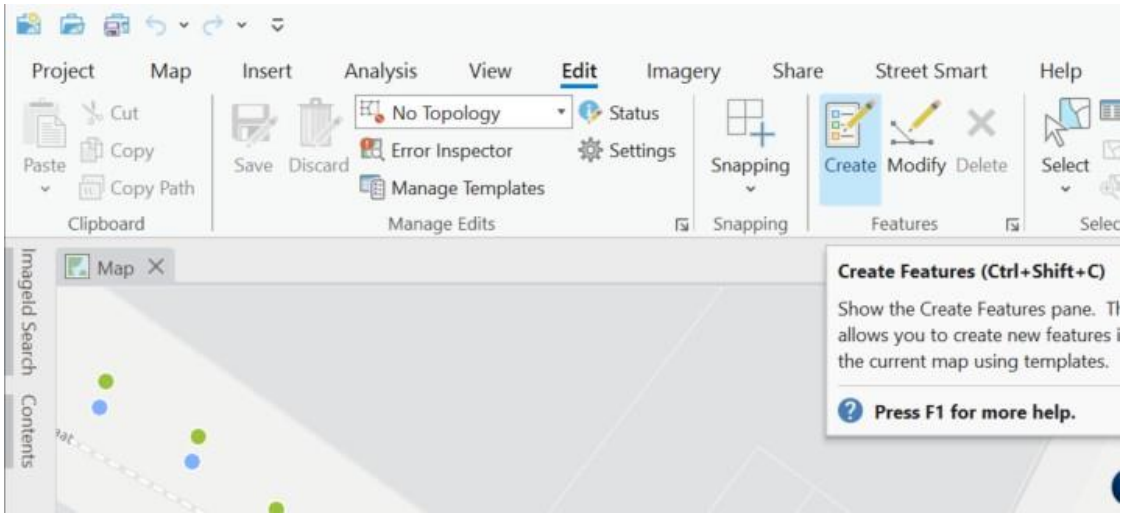
Measurements are stored in vector data layers. Point, line or surface measurements each are stored in separate point, line or polygon vector data layers. The following sections explain in detail how to perform the various types of measurements. To illustrate the principle of measuring, this manual will guide you

through the process of performing standard measurements first, before explaining measuring with Smart Click (which is turned on by default).

Point measurement

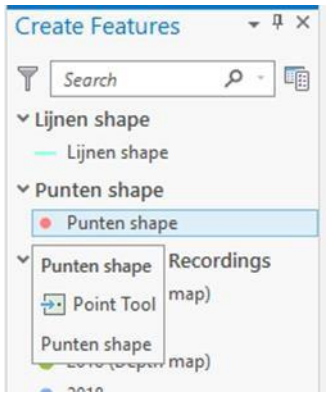
- ***Line / surface measurement Point measurement***

1. Add a point vector data layer to the map to store the measurement(s).
2. Open a Cyclorama in which the object to measure is clearly visible.
3. Select the tab: *Edit* ▶ *Create* to start editing mode in ArcGIS Pro.

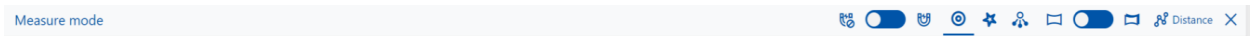


Start editing mode in the newly added vector layer.

4. Select the point layer in the *Create Features* window to start a point measurement.



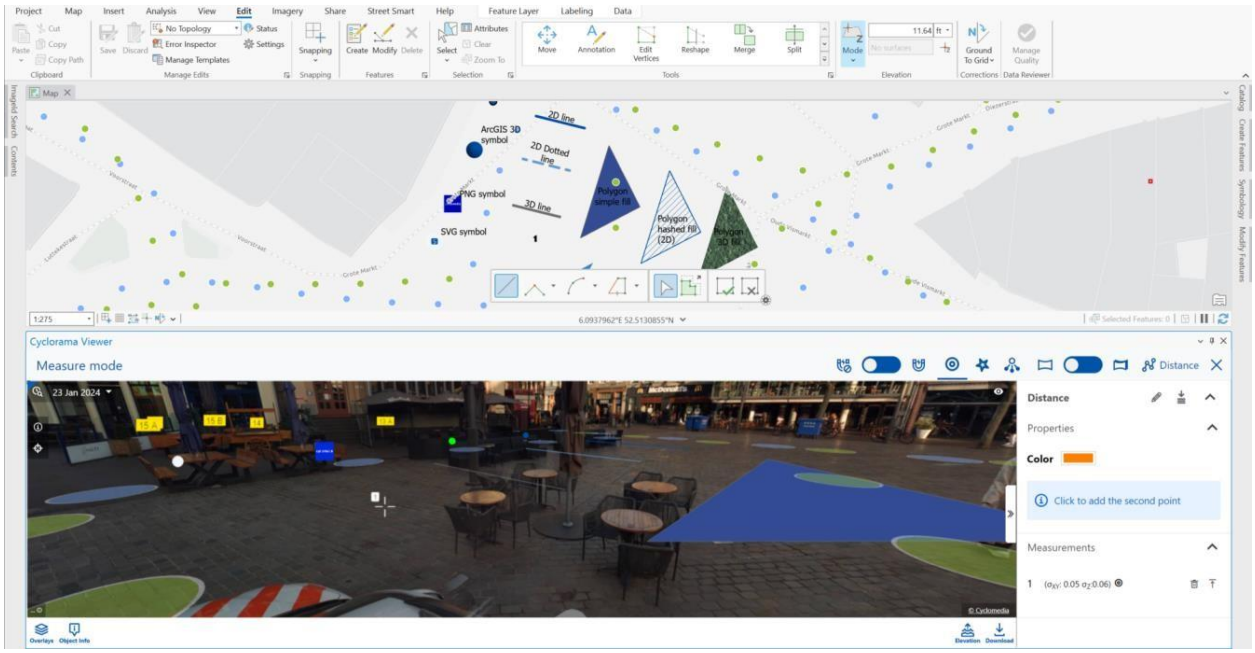
The Create Features window with several vector layers.

5. 

You can select the measure method in the Cyclorama (Measure Smart / Smart Click / Forward intersection)

The Measure mode bar

6. Select the point you want to measure in the Cyclorama.



Performing a point measurement.

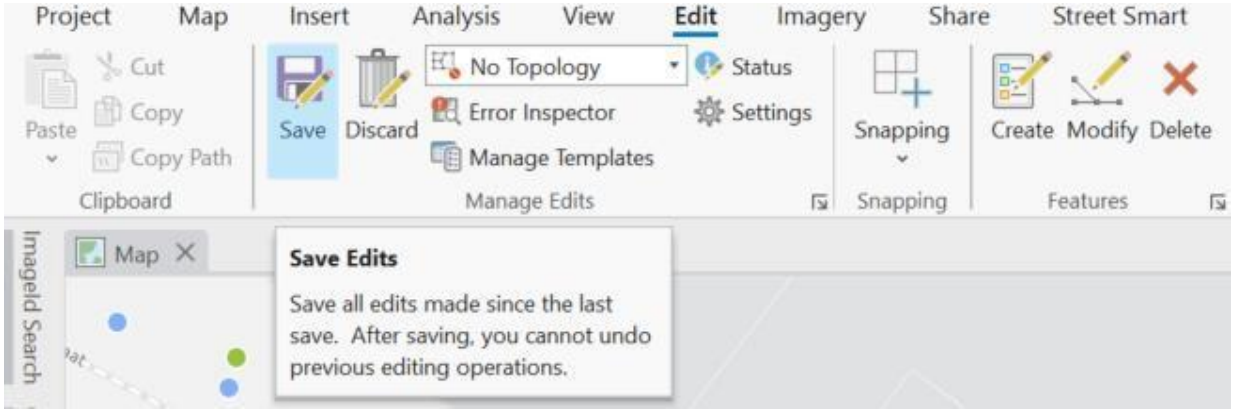
7. It is possible to change the measurement now. If the measurement is correct, you can proceed to point 8.

8. When you are finished with the measurement, click the <*Location*> button in the top of the

Measurement mode bar.

9. When you are finished with your measurements, save the edits to the vector layer in the

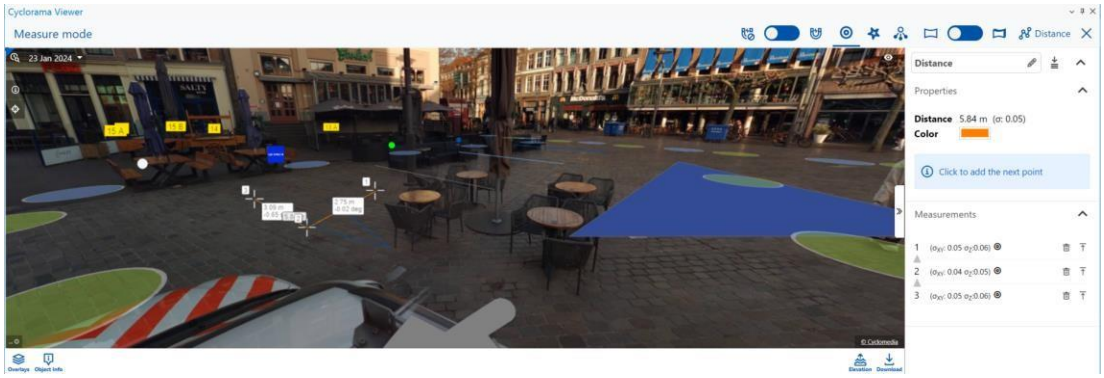
Edit tab in ArcGIS Pro, as shown in the figure below.



Save edits in ArcGIS Pro.

Line / surface measurement

1. Add a line or a polygon vector data layer to the map to store the measurement(s).
2. Open a Cyclorama in which the object to measure is clearly visible.
3. Select the tab: *Edit* ▶ *Create* to start editing mode in ArcGIS Pro.
4. Select the line or polygon layer in the *Create Features* window to start a line or a surface measurement.
5. You can select the measure method in the Cyclorama (Measure Smart / Smart Click / Forward intersection)
6. Select the point you want to measure in the Cyclorama.
7. Then select the second and subsequent points you want to measure, until you have selected all the points you want to measure.

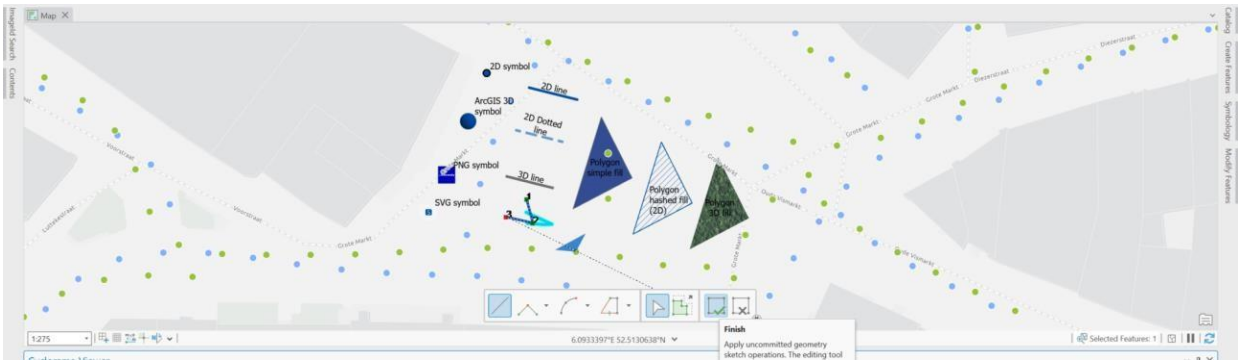


Select all the points in the first Cyclorama step by step.

8. It is now possible to edit points if this is desirable.
9. When you are finished with the measurement, click the <*Close*> button in the bottom of the

Selected Measurement Details window.

10. Click on the *Finish Sketch* button on the toolbar in the map or the <*Surface*> button in the measure bar in the cyclorama. With one of these buttons, you can close the measurement.



Use Finish Sketch on the toolbar of the bottom of the screen to close the measurement.

11. When you are finished with your measurements, save the edits to the vector layer in the *Edit* tab in ArcGIS Pro



After selecting the points in the second Cyclorama, the measurement is complete.

How to import a measurement in a Cyclorama

1. Select the *Move* or *Vertices* option in the *Edit* tab.



The Edit tab.

2. Select a feature on the map to select the overlay of the measurement in the Cyclorama.



Selected overlay in the Cyclorama.

3. If you want to edit the (selected) feature, select the *Vertices* option in the *Edit* tab.



Measurement shown in the Cyclorama (After use the Verticles option).

4. It is now possible to edit the measurement in the Cyclorama or on the map.

If you want to edit a measurement point in the Cyclorama, select that point in the map.

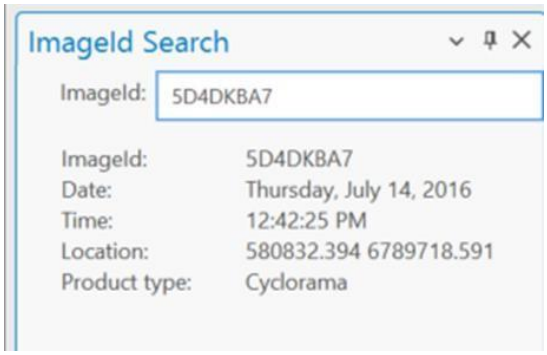
– 2.5 Cyclorama search functionality

1. To start the cyclorama search functionality, click on the *ImageId Search* button in the *Street Smart* tab.



ImageId search button

2. You will now see the ImageId search window.



Find Cyclorama window

3. Enter an imageId and you will get the results
4. Double-click the result and you go to the cyclorama.

Supported Overlay Symbols

Below is a list of the supported symbols in the overlays.

The following renderers are supported:

1. Single symbol renderer
2. Unique value renderer

The point Symbols:

- Supported:
- Circle symbols
- Color (including transparency)
- Stroke color (including transparency)
- Stroke width
- Picture marker symbols
  - Not supported:
- Size
  - ArcGIS 2D symbols:
- Supported:
  - Circle symbols
  - Color (including transparency)
  - Stroke color (including transparency)
  - Stroke width
  - Picture marker symbols
- Not supported:
  - All other styling and symbols
    - ArcGIS 3D symbols: not supported (the symbols are not displayed)

The Line Symbols:

- Supported:

- Color (including transparency)
  - Not supported:
- Line width
  - ArcGIS 2D symbols:
- Supported:
  - Color (including transparency)
- Not supported:
  - All other styling of the symbols
    - ArcGIS 3D symbols:
- Supported:
  - Color (including transparency)
- Not supported:
  - All other styling of the symbols

The Polygon Symbols:

- Supported:
- Color (including transparency)
  - Not supported:
- Outline
- Outline color
- Outline width
  - ArcGIS 2D symbols:
- Supported:
  - Color (including transparency)
- Not supported:
  - All other styling of the symbols
    - ArcGIS 3D symbols:
- Supported:

  - Color (including transparency)
- Not supported:
  - All other styling of the symbols

Troubleshooting & Hints

Some Issues users and administrators may encounter, together with the cause and the recommended solution or workaround for each. Where an issue depends on your organization's security configuration, the steps intended for IT and security staff are marked accordingly.

4.1 ArcGIS Pro closes unexpectedly when clicking the Street Smart tab (application-control / AppLocker environments)

Symptom On computers where application-control software (for example, Windows AppLocker) is enforced, ArcGIS Pro closes immediately when you click the Street Smart ribbon tab. In the Windows security event log this appears as Event ID 8004, with a message that CEFSHARP.BROWSERSUBPROCESS.EXE was prevented from running from a path inside the user's AppData folder.

Solution (for your security team): Add a narrowly scoped exception in AppLocker (or the equivalent application-control tool):

- **Recommended – path rule:** Allow the path
- %OSDRIVE%\Users*\AppData\Local\ESRI\ArcGISPro\AssemblyCache\{34D5B2FF-2566-4346-9C41-CB901D8811E1}* . The GUID is unique to the Street Smart add-in, so no other software can make use of this exception.
- **Alternative – publisher rule:** If you prefer rules based on a file's digital signature, create a publisher rule for CefSharp.BrowserSubprocess.exe.
- **If DLL rules are also enforced:** The same exception must also cover the DLLs in that folder (libcef.dll and the CefSharp.*.dll files); otherwise the viewer still fails after the executable is allowed.

Note Steps in AppLocker: Group Policy → Application Control Policies → AppLocker → Executable Rules → New Rule → Allow → Path condition → paste the path above → apply to the affected users. Neither your security policy nor the add-in is malfunctioning – this is an architectural interaction between Esri's add-in deployment model and standard security hardening, and the one-time exception resolves it permanently.

4.2 SSO sign-in page appears inside the Cyclorama Viewer panel and cannot be clicked

Symptom When you click "Log in with SSO", a separate small "Street Smart – Login" window should open and then close itself once you have signed in. On some computers this separate window does not open;

instead, your organization's sign-in page (for example, the Microsoft sign-in) loads inside the Cyclorama Viewer panel at the bottom of ArcGIS Pro, and the page appears frozen – it cannot be clicked.

How to confirm the cause (for your security team): Reproduce the issue and note the exact time, then check both of the following at that timestamp:

- Event Viewer → Applications and Services Logs → Microsoft → Windows → AppLocker → "EXE and DLL", for Event 8004 or 8003 involving CefSharp.BrowserSubprocess.exe.
- Your endpoint-protection product's logs, for a block involving ArcGISPro.exe or CefSharp.BrowserSubprocess.exe.

The definitive test is to perform the same sign-in on one machine with the security policy temporarily relaxed: if the separate Login window appears there, the environment is confirmed as the cause.

Permanent fix Apply the same security exception as in section 4.1 (the add-in's AssemblyCache folder with GUID {34D5B2FF-2566-4346-9C41-CB901D8811E1}). In endpoint-protection products, additionally trust ArcGISPro.exe launching CefSharp.BrowserSubprocess.exe. If you use a web proxy with SSL inspection, make sure login.cyclomedia.com and your organization's own sign-in domains are reachable.

4.3 Recording points not visible when using the "Human Geography" basemap

Symptom With the Esri "Human Geography" (or "Human Geography Dark") basemap active, the GeoCyclorama recording points do not appear on the map. Other basemaps (for example, Topographic) show them normally.

Cause The recording points are actually loaded correctly – you can confirm this by opening the recording points layer's attribute table, where all records are present. The cause is the structure of the Human Geography basemap itself. Esri deliberately builds it from three layers: a base layer at the bottom, plus a "Human Geography Detail" layer and a "Human Geography Label" layer that are configured as so-called reference layers. By ArcGIS Pro's design, basemap reference layers always draw on top of all data layers in the map, from any source, and the Detail layer includes a semi-transparent overlay intended to mute the data beneath it.

Small point symbols such as the recording points are effectively hidden underneath. This is documented Esri behavior and cannot be overridden by any add-in – ArcGIS Pro provides no mechanism for a data layer to draw above a basemap's reference layers. It affects any point layer from any vendor equally; you can verify this by placing one of your own point layers on this basemap, where it disappears the same way.

Solution (one click):

1. In the Contents pane, expand "Basemap" at the bottom of the layer list.

2. Uncheck “Human Geography Detail” and “Human Geography Label”.
3. The recording points appear immediately.

Alternatives Increase the transparency of the Detail layer (right-click → Properties → Transparency) to keep it while making the points visible, or use a different basemap (for example, Topographic or Light Gray Canvas) while working with Street Smart.